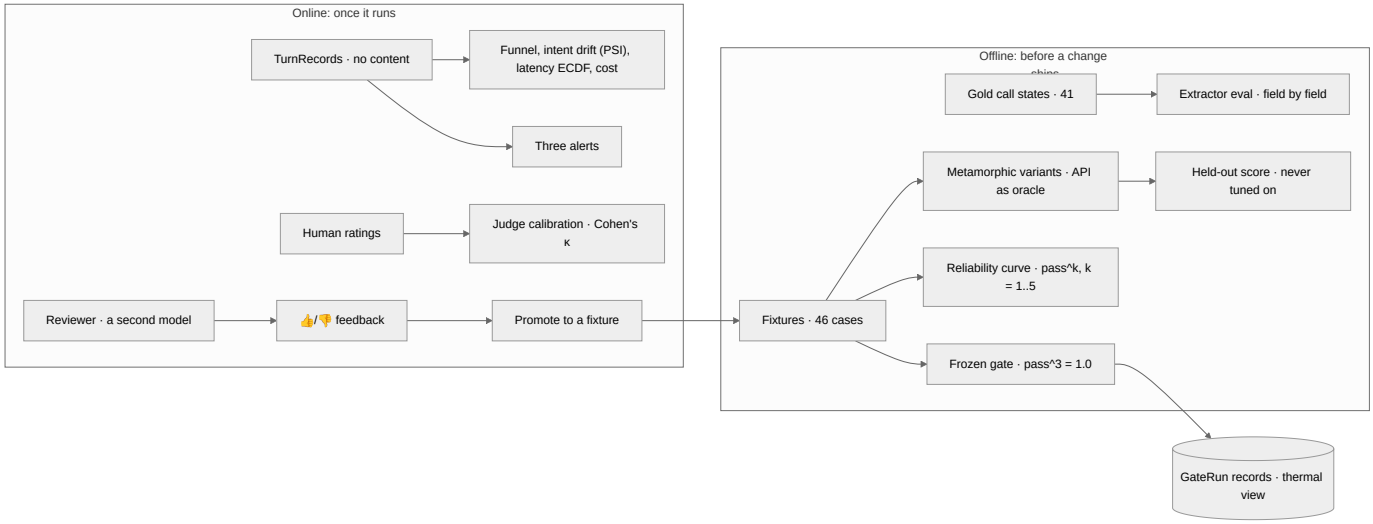
Data science: how the receptionist is measured

evaluation.md has the **results**. This page has the **methods**: what each number means, how it is computed, which code computes it, and how far to trust it. The decisions these methods carry out are summarised in the README's core technical decisions (8–13); the full reasoning is in SPEC §13.

The receptionist is a stochastic system: one model call a turn, at temperature 0, and gpt-4o-mini still varies between runs. The kit's judge, another model call, varies too. So no single run proves anything. Every claim here rests on repeated runs, a split of the failure's cause, and a record of every run.



1. The metrics at a glance

Metric	Question it answers	Definition	Code
Gate pass^3	Does every case pass every time?	a case passes when all 3 of 3 runs pass; the gate passes when every case does	`ops/gate_report.py`
Deterministic / judge pass^k	Was a failure ours or the judge's?	the same, counting only the checks in code (action, bookings, API invariants), or only the judge's verdicts	`ops/gate_report.py`
Extractor pass^k	Did the model hear the call right?	a fixture passes when all k extractions match its gold call state	`ops/extract_eval.py`
Field accuracy	Which part of understanding is weak?	per field, correct extractions ÷ compared extractions	`ops/extract_eval.py`
Reliability pass^k	What does a 3-run gate miss?	unbiased estimate of k passes in a row, from n = 5 runs	`ops/reliability.py`
Cohen's κ	Can the judge's verdicts be trusted?	judge–human and rater–rater agreement beyond chance	`ops/calibrate.py`
PSI	Have callers changed?	population stability index of the intent mix, last 7 days against the 7 before	`playground/analytics.py`

Metric	Question it answers	Definition	Code
p50 / p95 latency	How long is the silence?	nearest-rank percentiles of turns that called the model	`ops/gate_report.py`, `playground/analytics.py`
Cost per turn	What does a turn cost?	tokens × gpt-4o-mini prices ($0.15 in, $0.60 out per million)	`ops/gate_report.py`
Zero-call share	How often is the model not needed?	turns the turn gate settled ÷ all turns	`playground/analytics.py`
Held-out rules	Do the decisions hold on calls it was never tuned on, heard right?	variants passing the kit's checks and deciding as their source did, at zero tokens	`ops/variant_replay.py`
Held-out pass	How does it do on calls it was never tuned on?	generated variants passing all 3 of 3 runs, by the transform that made them	`ops/heldout.py`
Reviewer agreement	On real calls with no gold answer, how often does a second reader agree?	per field, turns where the agent's reading and an independent model's agree ÷ turns reviewed	`ops/review.py`

All paths are under `python/pirtis/`.

2. The data

Source	What it is	Size	Where
Fixtures	a call so far, the world (seed + setup) and the expected result: action, bookings, judge statements	14 given + 32 ours	`fixtures/`, `fixtures/extra/`
Seed	the saunas, customers and existing bookings every fixture starts from	1 file	`fixtures/seeds/base.json`
Gold call states	what the model should hear on each fixture it sees, field by field	41	`python/pirtis/agent/gold/`
Variant candidates	metamorphic copies of the given cases, checked by the kit's API	59 kept of 64	`fixtures/variants-candidates/`
Scenario cards	goals, facts and a twist for people calling the live app	15	`tests/human/scenarios.yaml`
Harness traces	one `result.json` per run, with every model call	per gate run	`python/runs/` (not in git)
Records we own	Session, Feedback, HumanRating, GateRun, Turn, Calibration	per call, per run	`python/var/` or the Azure file share
Raw reports	the output behind every number in the docs	per phase	`reports/`

Our 32 cases come in four kinds, each written for a different failure:

- **edge cases (20–29):** unknown callers, cancellations, a car too light, a weekend minimum, a correction mid-sentence
- **hidden-style (30–41):** the same rules with other callers, cars and dates, since the graders hold hidden cases "with the same rules"
- **red team (50–56):** prompt injection, a price demand, a request for another customer's details, a request for the system prompt

- **promoted (60–62):** real calls in the simulator that went wrong, turned into cases through the improvement loop (§9)

Gold states have their own rules: a gold file can `ignore` a field the fixture can't decide, list `one_of` groups where two readings are both right, and mark a field as compared by presence only (`heater_reason`: whether a reason was given matters, not its words). A test refuses a model-facing fixture without a gold state.

3. The frozen gate

Decided before any agent code was written (D-21) and never lowered.

- **Every case, 3 runs, all 3 must pass.** That is pass^3 = 1.0, not pass@3: a case that passes once in three tries is a failing case.
- **The verdict is split.** Each run's failures are sorted into the deterministic checks and the judge's. A judge-only failure is read before anything changes: our reply may be wrong, or the judge may have misread a correct one (§8).
- **Reported with every run:** cases, runs, both pass^k, model calls (and runs with none), tokens, cost, p50 and p95. Saved as a `GateRun` record, computed from the harness's own rows rather than passed in.

What a 3-run gate can and can't see. A case that fails a share f of its runs slips through a 3-of-3 gate with probability $(1 - f)^3$:

Per-run failure rate f	Gate catches it (3 runs)	Curve catches it (5 runs)
1%	3%	5%
5%	14%	23%
10%	27%	41%
20%	49%	67%
33%	70%	87%

So one passing gate says little about one case. It says more about the whole agent: **138 runs with no failure puts a 95% upper bound of about 2.1% on the per-run failure rate**, pooled over all cases (1 − 0.05^(1/138); the "rule of three" gives 3/138). That is why there is also a reliability curve.

Development set and held-out set

The 46 cases are the development set. Every change to the prompt and the policy was made while looking at them, so 46/46 is a development score: it shows the agent does what those cases ask, not how it does on calls nobody looked at.

A leak, found and closed (D-88). A scan of everything the model always sees (the system prompt and the schema's field descriptions) against every test case's caller lines found eight shared four-word runs. Two were the kit's own `turnaround` case word for word: "the wood smoke sets off my asthma" and "it has to be that one". An example copied from a test case teaches the model that one answer and flatters its score. The examples were rewritten in words no case uses, the heater one in the opposite direction ("electric heat gives me a headache"), and `agent/test_no_leak.py` now fails on any four-word run the prompt shares with a caller line in any suite, the generated variants included.

The held-out set (D-89): the 59 generated variants (§6). Code made them, the kit's API decided their right answers, and none was used to change the agent. `make heldout` runs them once, 3 runs each with all 3 needed, and reports them by transform:

Transform	Variants	What it tests
persona	16	another caller, another name and phone
shift	19	the same call one or two weeks later
noise	14	the caller's lines as speech-to-text writes them
tow	10	another car in the same towing band

Three layers answer "do the rules work beyond the cases?" (D-97), each separating one cause of failure:

Layer	What it tests	Tokens	Result
The rules on the held-out variants (`make heldout-rules`, in `make check`)	each variant run through the real agent with the caller **heard right** (the source's gold reading, moved by the same transform), against the kit's checks **and** a metamorphic relation: it must decide as its source did (the same kind of decision, the same saunas named, the same actions)	none	**59/59**. Control: a wrong reading is caught in 40 of the 43 variants it changes; the 3 misses are refusals whose right answer doesn't depend on the reading
A thousand generated calls (`agent/test_policy_fuzz.py`, in `make check`)	random groups, cars, sites, dates and sauna wishes against calendars made busier at random, straight into the policy: every offer is taken by the API as offered, nothing is written without a yes, only the caller's own bookings are read, and every refusal names its true reason	none	**1000/1000**: 400 offers, 475 refusals of all four kinds, 66 questions, 59 clashes with the caller's own booking. Control: with the rules copy switched off, it fails at call 4
The live held-out run (`make heldout`)	the same 59 variants with the real model and the kit's judge: hearing and wording, not only deciding	about $0.07	**58/59** on `1c0de71d1184` and again on the final `8612d72cfa25`: deterministic 59/59, judge 58/59. The one miss is the judge on `turnaround--noise` (the kit's hardest case, as speech-to-text writes it)

So the rules hold off the development set, and on the live run every decision on all 59 unseen calls was right; the one miss is how the judge graded one reply. Recorded as it came and not tuned on (`reports/after-16/heldout-rules.txt`, `heldout-1c0de71d1184.txt`).

The protocol: score a batch **once, before tuning anything on it**, and report the number as it comes. A variant used to fix something is development data from then on; the next held-out score needs a fresh batch (`make variants`).

What closing it showed. The gate on `09302402ead3`, the first agent without the leak, failed 45/46 (deterministic 46/46, judge 45/46). `turnaround`, whose words were the main leak, still passed 3/3. Case 30 did not: its caller asks "how much is the electric six seater per day?", which had been a prompt example word for word. Without it the model recorded no sauna, so the reply gave the whole price list

and the judge's "gives the daily price of Vidutinė" failed. A zero-token replay reproduced the reply exactly. The leak had been carrying that case; code now reads a price question for its sauna (D-91), and the unit tests use wordings no case contains (`reports/after-16/`).

The extractor eval on that fix (`6c6118d087e8`) passed case 30 3/3 and found one more: 40/41, with `barge-in-correction` recording a heater preference in 2 of 3 runs that the caller never stated, taken from our own cut-off offer of an electric sauna. A heater preference now counts only from a caller who talked about heating (D-92).

Status: the numbers in §3–§5 belong to agent `bd98c5cf8006`, before the leak was closed. `1c0de71d1184` added the fixes from the first manual test session (D-93 to D-96), passed the gate and scored 58/59 held out; the current agent, `8612d72cfa25`, adds the thread of the call (D-98 to D-101) and passed the gate, 46/46 cases at 3 of 3 (p95 2.19 s), and scored 58/59 held out, the same as before (`reports/after-16/heldout-8612d72cfa25.txt`); its extractor eval is not recorded yet.

4. The reliability curve: pass^k

Every case runs n = 5 times. For a case with c passes, the unbiased estimate of passing k runs in a row is

```
pass^k = C(c, k) / C(n, k)
```

the estimator τ-bench uses. A suite's pass^k is the mean over its cases; k = 1 is the plain pass rate. A flat curve means the agent is consistent; a falling curve means a single good run hides flakiness.

Worked example, case 62 ("eight of us, maybe nine"), c = 4 of n = 5:

- pass^1 = 4/5 = 0.8
- pass^3 = C(4,3)/C(5,3) = 4/10 = 0.4, so it would **fail a 3-of-3 gate 60% of the time**, even though it had passed two gates already
- pass^5 = 0

Averaged with 45 cases at 1.0, that one case gives the whole curve: 0.9957 at k = 1 down to 0.9783 at k = 5. The cause was found in one read of the traces (the model took the lower number of a hedge), fixed in code (D-79), and the four hedge-sensitive cases then passed 20/20.

Run it with `make reliability` (about $0.30 with the judge).

5. The extractor eval: understanding, isolated

End-to-end failures mix three causes: the model misheard, the policy decided wrong, or the judge misjudged. The extractor eval isolates the first (D-36).

- For each of the 41 gold states, the real extraction runs **k = 3 times** against the live model.
- Each run is compared field by field: names normalised and case-insensitive, presence fields by presence, `one_of` groups by any alternative.
- A fixture passes only if all k runs match. Per field, accuracy = correct ÷ compared.
- It exits non-zero on any miss, so it can gate too.

Run it with `make eval-extract`. The gate job runs it on the nightly suite as well.

How understanding was engineered, measured at each step

The extraction is one strict-JSON call into a `CallState` (`agent/understand.py`). The schema and its validators are where most of the gain came from, not the prompt:

Technique	Why	Effect measured
Evidence before value. Field order is the order the model writes: a gist, then each caller quote before the value taken from it, consent last	the model commits to the words before the number	`heater_reason` before `heater_pref` fixed an asthma case read as a wish for wood (D-69): 22/23 → 23/23
Quotes verified in code. A tow figure, a sauna and consent count only if their quote is in the caller's own lines (≥ 70% of its words in one line), with a number for a tow figure, and consent from the latest turn only	the model picked saunas from our own lines and set consent with no offer made	12/20 → **20/20** (D-54); prompt rules alone had made consent worse (80%)
Asymmetric hedges. "Eight, maybe nine" people → the higher; "two or three tonnes" → the lower	under doubt, choose the reading that can never produce an offer the API rejects	the case-62 flake above (D-79)
Consent voided by an attack. A "yes" that names its own price or talks to the system is not a yes	prompt injection through the consent field	red-team case 50: "book it at zero euros" had been taken as a yes (D-74)
Plausibility validators. Dates must parse, group sizes and tow figures must be plausible, a sauna must exist	a wrong guess should cost a question, never a booking	—

Field accuracy, final run: all 15 fields at 100% over 123 extractions (`reports/phase-16/eval.txt`).

6. Metamorphic variants: the API as the oracle

The hidden cases have "the same rules, other callers, cars and dates". Variants make exactly that from the 14 given cases (`ops/variants.py`, D-76):

Transform	What changes	What must stay true
`persona`	another seed customer with no bookings calls	the expected outcome
`shift7`, `shift14`	the whole call is one or two weeks later: today, every spoken and written date, setup and expectation	weekdays stay right; the same saunas fit
`noise`	the caller's lines as speech-to-text writes them: fillers, a restart, lower case, figures as digits	the expected outcome
`tow`	the car tows another figure in the same band	the same saunas fit

The oracle is the kit, not our code. A variant is kept only if the kit's loader reads it, the kit's API accepts every booking it expects in a fresh world, and the API returns the same fitting saunas as for the source. The caller's needs come from the source's gold state, moved by the same transform. Of 64 generated, **59 were kept**; 5 were dropped because a date shift changed which saunas fit. They wait in `variants-candidates/` for a person's review (`make variants-accept`) before they join the gate.

7. The human evaluation and judge calibration

The judge decides half of every gate verdict, so how far it agrees with people is measured, not assumed (D-47, D-77).

- **Protocol:** people call the live app with one of 15 scenario cards and rate the call: goal reached (yes/no), and four items from 1 to 5: natural, numbers clear, one question at a time, turn-taking (tests/human/README.md).
- **Binarising:** a score of 4 or 5 counts as "yes".
- **The judge's side:** each rubric item becomes a statement for the kit's own judge, asked of every receptionist reply in the call; the call passes the item only if every reply does. The goal is asked once, on the last reply, as the card's success sentence.
- **Agreement:** per item and pooled, the share of matching verdicts and Cohen's κ:

  ```
  κ = (p_o − p_e) / (1 − p_e),   p_e = p_a·p_b + (1 − p_a)(1 − p_b)
  ```

  κ is reported as undefined when p_e = 1 (both raters constant), not as 0 or 1.
- **Rater–rater κ:** where two people rated the same call, the same statistic between them, as the ceiling the judge can be held to.
- **Outcome from the record:** the Human eval view also checks each call's own record (booked, moved, cancelled, transferred) against the card's outcome, beside what the rater said.

Status: built and tested, not yet run: there are no ratings yet. `make calibrate` refuses to save an empty calibration, so the console can't show one as a result.

8. Judge failure analysis

A judge-only failure is read, never rerun until green. The patterns found, each with the evidence in `reports/`:

- **Polarity:** a negative criterion ("does not say whether the booking is still on") marked failed with a reason that agrees with the reply. Fixed by replies that state the fact outright (D-51, D-87).
- **"Or" read as "and":** "asks what car, or how much it can tow" failed a reply that asked one. Replies now cover both readings (D-51).
- **No date, no literal grading:** the judge isn't told today's date and grades against the caller's request, so our statements stay within what it can check; the rest is checked by gold states and unit tests (D-72).
- **Rate of the remaining flake:** fixture 01, 1 judge failure in about 43 runs on an identical reply, before D-87; 20/20 after.

9. Online: telemetry, monitoring and drift

One `TurnRecord` per turn, with no content (`agent/telemetry.py`): which path the turn took, the intent, the decision, the actions, model calls, tokens, latency, every API call as method, status and code, whether it fell back, and the phone as a salted hash. No text, no names. It renders to OpenTelemetry GenAI attributes through one function. Every analytics view reads only these records, gate runs or the seed, so none of them can name a customer.

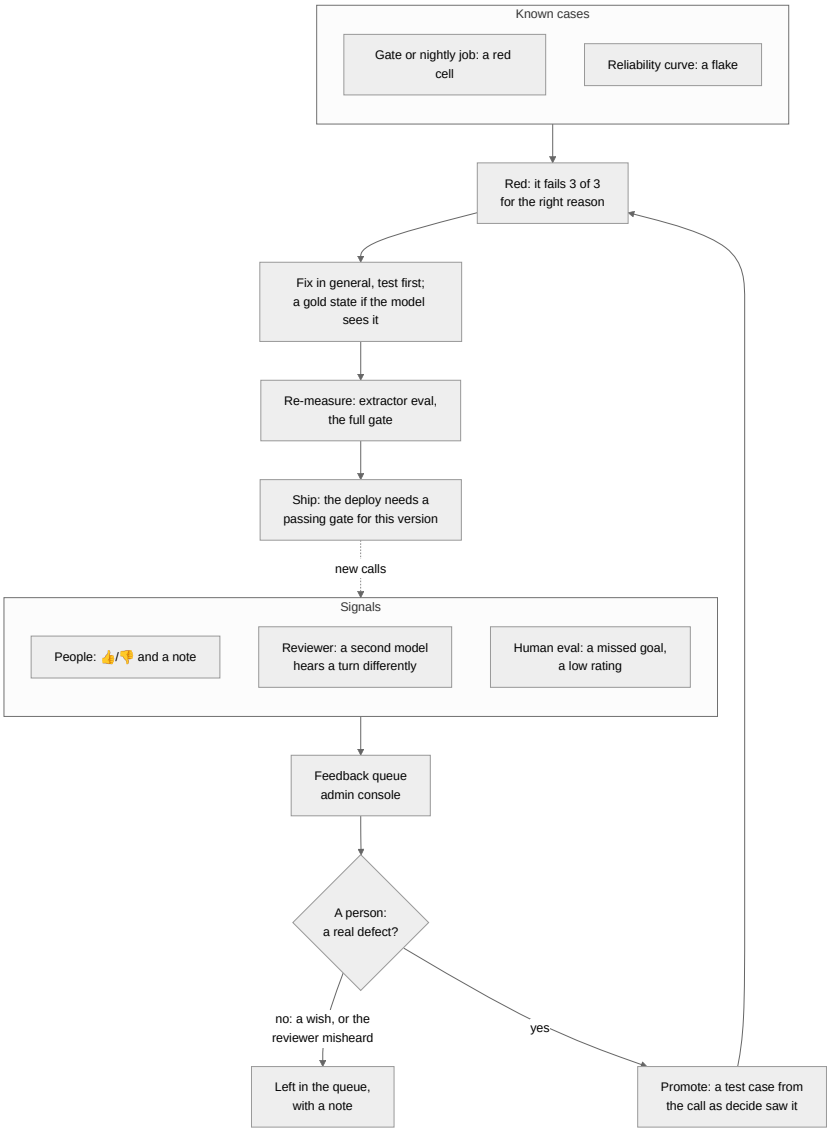
Admin view	Analysis
Conversations	the funnel (started → identified → offered → booked) and the exits (transferred, hung up, fell back); the intent mix against the previous window; the latency ECDF; tokens per model turn; zero-call share; cost; API calls by status
Intent drift	PSI = Σ (p − q) · ln(p / q) over intents, shares floored at 10^{-4}; 0.10 warns, 0.25 alerts (D-43). Below 30 turns in either window it says "too few" rather than a number
Reliability	the thermal view: a row per case, a column per run grouped by agent version; each cell pass, fail, or judge-only fail
Fleet / what if	each sauna's next 42 days (free, booked, turnaround, blocked), and for a hypothetical booking, each rule it breaks, in words, from the policy's own rule mirror
Human eval	per card: ratings, raters, goal rate, the record's outcome rate, rubric means; the rubric's spread; the latest calibration

Latency is measured per `decide()` call, model turns only; turns the gate settles take 0 ms and are counted apart, so they can't flatter the percentiles.

Alerts (Log Analytics, every 15 minutes over the last hour): a gate regression, any fallback or API error, and p95 above 3 s. Gate and replay turns never reach the log, so fault cases made on purpose can't fire them.

10. The feedback loop

How what goes wrong becomes a permanent test, and how the loop is kept from fooling itself. Built in `playground/promote.py`, `ops/review.py` and the admin console's Feedback view (SPEC §11).



The signals

Signal	Catches	Cost	Status
People's 👍/👎 on any reply, or *Flag this turn* in a trace	what a person notices: a wrong answer, a clumsy line, a missed cue	testers' time	built and used: cases 60–62
Manual test sessions in the Pirtis Test Log: 34 cases marked pass, fail, blocked or skipped, plus findings	what a scripted walk through the product finds: recognition, rigid rules, missing affordances	testers' time	first session 1 Nov: 24/34, logged in `reports/manual/`
The reviewer (`make review`): another model re-reads each model turn and compares its reading with the agent's, fact by fact	misheard facts on real calls, which have no gold answer	one call to the reviewer's model per model turn	built and tested; not yet run on live calls
Human evaluation with scenario cards	how it sounds to people, and how far to trust the judge	testers' time	built, not yet run
The gate and the nightly job	a regression on a known case	about $0.05 for a full gate	running nightly in Azure
The reliability curve	a flake three runs can miss	about $0.30	run once (Phase 13)

The rules that keep it honest

1. **A person decides.** Nothing becomes a test case on its own (D-30). A reviewer flag is a suggestion with both readings shown, never a label, and it never overwrites a person's own rating.
2. **Seed people only** reach git: a promoted case carries no real name, phone or address (D-39, D-66).
3. **Red before green.** A promoted case must fail on the agent that got it wrong, for the right reason, or it doesn't test the defect yet; its judge statement is sharpened.
4. **Fix the rule, not the case.** No case's words in the prompt (`test_no_leak.py`); the booking rules come from the API and are tested against it, not against the cases.
5. **Every fix is re-measured.** A changed prompt or policy is a new `agent_version` (D-56), so it needs the extractor eval and the full gate before the deploy lets it out (D-85).
6. **The held-out set stays out** of the loop until it has been scored (§3); a variant used to fix something is development data afterwards.

What has come through it

Case or change	Signal	What it found	Fixed by
60 books-didzioji-for-six-with-no-mains-power	👍 on a live call	a regression guard for a four-turn booking without power; it entered green, and showed the gold set had silently stopped covering new cases	D-70: every model-facing case needs a gold state, checked against every suite
61 introduces-herself-and-is-helped	👎	"Hi, it's Rasa Paulauskienė" refused as a question about another customer	D-67: a greeting intent
62 nine-people-hear-the-largest-sauna-seats-eight	👎	nine people asked about their car before hearing no sauna seats nine; later, on the curve, a 1-in-5 flake	D-68, then D-79 (§4)
fixture 01, final gate	a red cell	the judge flipped on a correct reply, 1 run in about 43	D-87: the reply states the fact outright
the prompt's examples	a review of the method	test-case wording in the prompt	D-88 (§3)
30 price-before-identification	a red cell: the gate on `09302402ead3`, on the judge	once its words left the prompt, the model recorded no sauna for a price question, and the caller got the whole price list	D-91: code reads a price question for its sauna
manual session GS 2, 1 Nov (25 of 36 marked)	people: C1, C2, C8 failed; notes on C3b, C4b	no closure after a booking, a refusal repeated forever, the thread lost after a check-in, "book it" not taken as a yes, "what's free?" met with a question	D-98, D-99: the agent reads its own last line
manual session GS, 1 Nov (`reports/manual/`)	people: 9 of 34 cases failed, 1 finding	names an English recogniser respelled ("Thomas"); ten people asked for a name first; "which dates are free?" read as a price; a two-part tow question; real quiet not treated as silence; hidden 👍/👎	D-93 to D-96 in the agent; the simulator and console in code
27 barge-in-correction	the extractor eval on `6c6118d087e8`, 2 of 3 runs	a heater preference the caller never stated, taken from our cut-off offer of an electric sauna	D-92: a heater preference needs a caller who talked about heating

Measuring the loop itself

- **Reviewer agreement per field**, printed on every `make review`: the nearest thing to an accuracy figure for real calls. A field whose agreement drops is the place to look first.
- **Not built yet:** the share of reviewer flags a person confirms (the reviewer's precision), and the time from a flag to a fix. Both need the reviewer to run on real calls first.

11. Data governance

- **Six versioned record types** (`ops/model.py`): frozen pydantic models, unknown fields refused, time zones required, each with a retention period: 30 days for calls and feedback, 90 for turn records, a year for ratings, gate runs and calibrations.
- **A privacy class on every field:** personal, pseudonymous or none. A test pins the personal fields to exactly the call transcripts, bookings, steps and feedback notes.
- **Atomic, index-free storage:** one JSON file per record, written by rename, so the playground and the gate job can write at once (`ops/store.py`).

Full table: SPEC §9b; where personal data crosses a boundary: architecture/data-flow.md.

12. Limits of these numbers

- **Small n.** 46 cases × 3 runs, or × 5 for the curve. A per-case rate is coarse; the pooled bound in §3 is the stronger statement, and it assumes cases are alike.
- **The curve is one version behind.** It was measured on `a40b5e6fcd67`; D-79 and D-87 followed, each re-measured on the cases they touch (20/20), not on a new full curve.
- **The judge is a model too.** Its verdicts aren't calibrated against people until the human evaluation runs.
- **No confidence intervals** are printed by the gate report; §3 gives the arithmetic.
- **Cost is the agent's only;** the judge's calls aren't in the traces.
- **Latency is `decide()` time,** not the caller's: speech-to-text and text-to-speech would add to it.
- **The gate is the development set** (§3). Off it: the rules hold on all 59 held-out variants and a thousand generated calls at zero tokens, and the live held-out run scored 58/59 (every decision right, one judge miss). One batch of 59, run once; the reviewer is built but not yet run on real calls.
- **The current agent is only partly measured.** The numbers in §3–§5 are for `bd98c5cf8006`. `09302402ead3` closed the prompt leak and failed the gate 45/46; `6c6118d087e8` fixed that case and scored 40/41 on the extractor eval (§3); `239ddaf571d0` fixed that one; `1c0de71d1184` passed the gate and scored 58/59 held out; the final `8612d72cfa25` passed the gate (46/46) and scored 58/59 held out; its extractor eval is still to be recorded.
- **Drift is watched on intents only,** without an alert.
- **Fixtures are hand-written or promoted from the simulator,** not from real phone calls.

13. Reproduce

From `python/`, with a key in `.env`:

Command	What it does	Model spend
`make check`	lint, strict types, tests, including every metric's unit tests	none
`make gate`	the frozen gate and its report	about $0.05 + the judge
`make eval-extract`	the extractor eval, 3 runs per gold state	about $0.05
`make reliability`	pass^k for k = 1..5	about $0.30 with the judge
`make variants`	generate and check the variant candidates	none
`make heldout-rules`	the 59 variants through the real agent, heard right, at zero tokens (also in `make check`)	none
`make heldout`	the 59 generated variants once, by transform: the held-out score	about $0.07 + the judge
`make review`	a second model re-reads the last day's calls and queues disagreements; `DRY=1` only reports	one reviewer-model call per model turn
`make calibrate`	judge–human agreement on saved ratings	one judge call per item per reply
`python -m pirtis.ops.gate_report --dir runs/gate`	the report alone, from saved traces	none